Kemal Ozturk

SOFTWARE ENGINEER · PRODUCT DEVELOPMENT & SYSTEMS DESIGN

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PROFESSIONAL SUMMARY

Software engineer with 7 years of experience designing, shipping, and operating software across web, desktop, and backend services. Leads projects from technical design through production rollout, with experience in reusable architecture, authorization, concurrency, and reliability. Maintains open-source software spanning Rust services, Linux/Windows hardware integration, and web applications.

TECHNICAL SKILLS

Product: TypeScript, JavaScript, React, Angular, Next.js, Svelte, Redux/Redux-Saga, Node.js, HTML, CSS/SCSS

Platform: REST APIs, Postgres, Firebase, GCP, AWS, S3, Temporal, Cloudflare, Sentry, Grafana, LaunchDarkly

Systems: Rust, Tokio, Poem/OpenAPI, Python, C, C++, Linux, Nix/NixOS, Arch Linux

Media & real time: WebRTC, Daily, MediaRecorder, WebCodecs, WebAssembly, Web Workers, Electron

Development tools: GitHub Actions, Nx, Turborepo, Vite, Retool, Cursor, Claude Code, OpenAI Codex, Zed

PROFESSIONAL EXPERIENCE

Software Engineer — **June 2023 – September 2026**

Descript

- Led Scheduled Rooms from architecture RFC and phased implementation through full rollout, coordinating changes across APIs, permissions, and recording clients. Adoption reached 500+ weekly creations and roughly 25% of Rooms recordings.
- Designed invite-scoped authorization with server-enforced permissions, project-scoped credentials, and isolated sessions, enabling guests to record without project membership. Coordinated security review and staged releases, including hostless recording to general availability.
- Architected the initial Web Recorder engine in TypeScript with Redux-Saga orchestration, separating capture logic from the UI. Reused across Editor, Media Library, Quick Recorder, and Rooms for multi-track camera capture at 4K/30 FPS, microphone, screen, system audio, and device switching.
- Delivered instant playback so users could review takes while recording chunks were still uploading. Built a Web Worker upload manager with progressive uploads and progress-aware timeouts; resolved asynchronous queue races and implemented in-call message serialization and coalescing.
- Shipped recording visualization, cancel/restart, and pre-roll/live playback; improved A/V synchronization from approximately 700 ms to 50 ms. Built an internal RNNoise-backed WebAssembly denoise prototype and profiled capture performance to improve frame rate and bitrate.
- Owned production incident response and recording recovery across web and desktop. Built analytics dashboards, alerts, memory telemetry, deployment safeguards, and Support diagnostics; investigated device failures, authorization errors, and rate-limit issues.

Software Engineer
January 2022 – August 2023

SquadCast (acquired by Descript)

- Built core remote-recording product features using Angular, TypeScript, Next.js, Firebase, and GCP, spanning capture, authentication, account lifecycle, third-party integrations, and onboarding. Migrated selected microservices to serverless cloud functions to reduce costs and support scaling.
- Designed and delivered Canvas and SquadShots features, app-wide dark mode, mobile-first layouts, and accessibility improvements. Profiled CPU usage and fixed memory leaks across Studio, Chat, Dashboard, and audio metering.
- Created Nx executors and generators that reduced feature scaffolding time from 30 minutes to 5, standardizing setup for new features.

Software Engineer
August 2019 – September 2021

Peel9

- Delivered records-management features used by 10+ police departments around Cincinnati, including incident approval, traffic-crash diagrams, modular forms, and account recovery, using Node.js, Express, and Handlebars.
- Led code reviews and mentored junior engineers; maintained the GitHub repository and built a Leaflet plugin for dynamic crash-map markers.

SELECTED PROJECTS

Framework Control
2026 – Present

- Built and maintains a Rust service (Tokio, Poem/OpenAPI) and Svelte UI for Framework laptops and desktops on Linux and Windows. Implemented telemetry, fan curves with hysteresis and rate limits, and battery/power controls through the Framework CLI and platform-specific interfaces.
- Shipped Windows MSI and Linux systemd/udev support, with AUR, nixpkgs, and NixOS distribution. Added tests for fan-control behavior and CLI parsing; solo-maintained project reached 220+ GitHub stars and 7 forks.

Cihan Radyo
Volunteer · Ongoing

- Built and maintains a Svelte/TypeScript radio and podcast PWA with local playback state, lock-screen controls, Turkish-aware search, and Google Drive sync. Released on Google Play and the iOS App Store, using a TWA wrapper on Android.
- Maintains the publishing pipeline from audio folders to podcast RSS feeds. An August 2026 Cloudflare snapshot recorded approximately 1.4K daily unique visitors, 334K requests, and 20 GB served in a day, with 61% cached.

RGBPiano
2023 – Present

- Built a MIDI-driven piano LED visualizer: a TypeScript host computes note mapping and envelopes, then streams binary frames over WebSocket to a Python service on a Raspberry Pi Zero driving WS2812 LEDs. A Svelte UI previews and configures the visual behavior.

EDUCATION

Bachelor of Science, Information Technology
2021 · GPA 3.97/4.0

University of Cincinnati

Coursework toward a Bachelor of Science in Computer Science
2016 – 2018 · GPA 3.83/4.0

Ankara University